Intermediate Microeconomics

Week 3 · Day 1 — Duopoly: Bertrand, Cournot, Stackelberg

Summer School

One market, three ways two firms can compete — and where each lands between monopoly and perfect competition.

Where we are — Week 3 plan

So far a market has had either *many* price-taking firms (perfect competition) or *one* (monopoly). This week we live in between: a handful of firms who each feel their rival.

The week at a glance

Day 1 Duopoly: Bertrand, Cournot, Stackelberg — and the welfare ranking
Day 2 Game theory: how to think about strategic choice in general
Day 3 Repeated games: can rivals sustain cooperation over time?
Day 4 Externalities, public goods, the commons

Our running market — all day

Inverse demand $P = 120 - Q$; each firm produces at constant marginal cost $c = 30$. We solve it four ways and line the answers up.

Learning goals — Day 1

By the end of today you should be able to:

1. Recall the two benchmarks — perfect competition and monopoly — as prices, quantities and welfare.
2. Solve a **Bertrand** (price-setting) duopoly and explain why two firms can be enough for a competitive outcome.
3. Solve a **Cournot** (quantity-setting) duopoly using best-response (reaction) functions.
4. Solve a **Stackelberg** (sequential quantity) duopoly by working backwards, and identify the first-mover advantage.
5. Rank all outcomes on one picture and compute the deadweight loss of each.

A note: today we find each "resting point" by asking what each firm does when it best-responds to the other. Next week we give that idea its proper name.

Setup and the two benchmarks

Market inverse demand and cost: $P = 120 - Q, \qquad MC = c = 30.$

Perfect competition — price equals marginal cost

$$P = 30 \;\Rightarrow\; 30 = 120 - Q \;\Rightarrow\; \boxed{Q_c = 90}, \qquad \text{industry profit} = 0.$$

This is the *efficient* quantity.

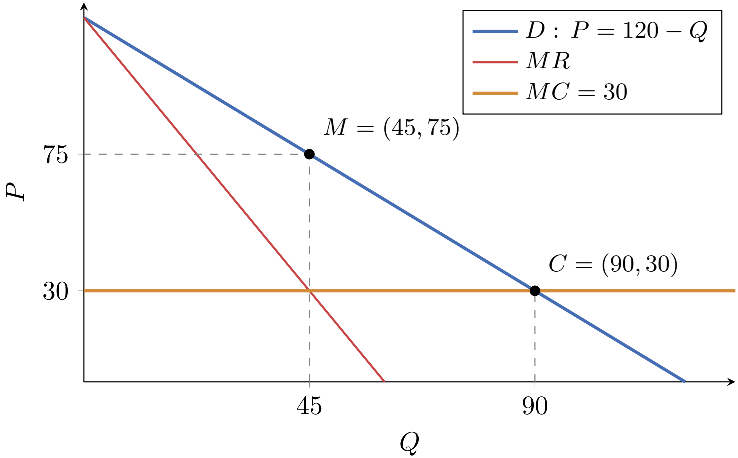
Monopoly — marginal revenue equals marginal cost

$$MR = 120 - 2Q = 30 \;\Rightarrow\; \boxed{Q_m = 45}, \quad P_m = 75, \quad \pi_m = (75 - 30) \cdot 45 = 2025.$$

Keep this in mind

Every duopoly outcome today will sit *between* monopoly $(45, 75)$ and competition $(90, 30)$ on this demand line.

The two benchmarks — the picture



Monopoly M restricts output and charges a high price; competition C pushes price down to marginal cost.

Welfare of the two benchmarks

Welfare = area between demand and MC up to Q: $W(Q) = \int_0^Q (120 - x)\,dx - 30Q = 90Q - \frac{1}{2}Q^2$.

Consumer surplus, producer surplus, total

	Q	P	$CS = \frac{1}{2}Q(120-P)$	$PS = (P-30)Q$	W
Perfect competition	90	30	4050	0	**4050**
Monopoly	45	75	1012.5	2025	**3037.5**

- Competition delivers the largest total surplus, $W = 4050$; all of it goes to consumers.
- Monopoly restricts output to 45 and destroys $DWL_m = 4050 - 3037.5 = 1012.5$ of surplus — the classic welfare triangle.

Today's question

How far toward competition does a *second firm* pull us? The answer depends on *how* the two firms compete.

Bertrand — firms set prices, consumers buy the cheapest

Two firms, identical product, same constant cost $c = 30$. Each *posts a price*; the lower price wins the whole market; ties split it.

The undercutting logic

- Suppose both post $P = 75$ (the monopoly price), sharing $\pi = 2025$, so 1012.5 each.
- Firm 1 posts 74 instead: it steals the *entire* market and earns almost the whole monopoly pie.
- So firm 2 undercuts to 73, firm 1 to 72, …each wants to be just below the other.

Where does undercutting stop?

The only price no one wishes to undercut is $P = c = 30$: cutting below cost means losing money. The market rests there.

The Bertrand outcome: two firms are "enough"

Result

$$P_B = c = 30, \qquad Q_B = 120 - 30 = 90, \qquad \pi_B = 0.$$

Exactly the *perfectly competitive* outcome — with only **two** firms.

The Bertrand paradox

Homogeneous product + price competition + equal constant costs ⇒ price collapses to marginal cost. Two is as good as a thousand.

What breaks the paradox in real markets

- capacity limits (a firm cannot serve the whole market);
- differentiated products; or costs that differ across firms.

Then price sits *above* c — which is exactly why Cournot (next) is often the more realistic model.

Cournot — firms choose quantities simultaneously

Now firms pick *output*, not price. Each chooses q_i; the market price is whatever clears total supply:

$$P = 120 - (q_1 + q_2).$$

The strategic catch

My price — and my profit — depends on how much *you* produce. I must form a best guess of q_2 and choose my best reply.

Firm 1's problem (firm 2 is symmetric)

$$\max_{q_1} \ \pi_1 = \left[\, 120 - (q_1 + q_2) \,\right] q_1 - 30\, q_1 .$$

The rival's output q_2 shifts firm 1's *residual demand* inward: the more you make, the less room is left for me.

Best-response (reaction) functions

Firm 1's first-order condition

$$\frac{\partial \pi_1}{\partial q_1} = 120 - 2q_1 - q_2 - 30 = 0 \implies \boxed{q_1 = \tfrac{90-q_2}{2}}.$$

This is firm 1's **best response**: its profit-maximising output for any belief about q_2. By symmetry, $q_2 = \frac{90-q_1}{2}$.

Reading the reaction function

- If the rival makes nothing ($q_2 = 0$): $q_1 = 45$ — I act as a monopolist.
- If the rival floods the market ($q_2 = 90$): $q_1 = 0$ — no room left.
- Each extra unit the rival makes, I cut my own output by half a unit.

Key idea

Outputs are *strategic substitutes*: more of yours makes me want less of mine.

Where the two best responses meet — the algebra

A resting point needs *both* firms best-responding at once. Solve the pair:

$$q_1 = \tfrac{90-q_2}{2}, \qquad q_2 = \tfrac{90-q_1}{2}.$$

Substitute the second into the first

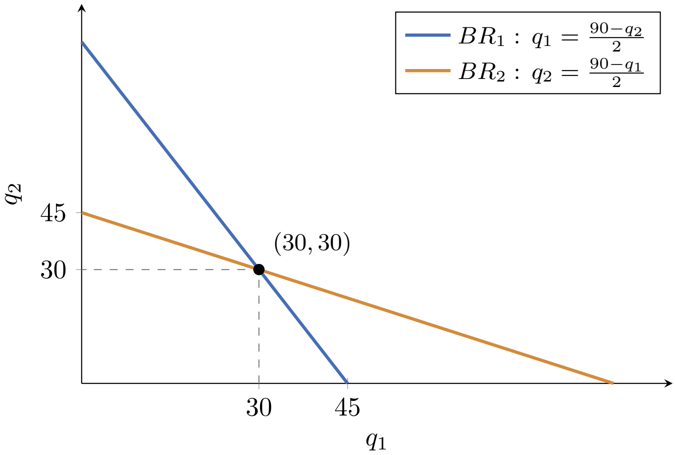
$$q_1 = \frac{90 - \frac{90-q_1}{2}}{2} = \frac{90}{2} - \frac{90-q_1}{4} \implies 4q_1 = 180 - 90 + q_1 \implies 3q_1 = 90.$$

Cournot outcome

$$\boxed{q_1 = q_2 = 30}, \qquad Q_{Cou} = 60, \qquad P_{Cou} = 120 - 60 = 60.$$

Numeric check: best reply to q_2=30 *is* $\frac{90-30}{2} = 30$. ✓ *Neither firm wants to move.*

Where the two best responses meet — the picture



The two reaction curves cross exactly once, at $(30, 30)$ — the only pair where each firm is best-responding to the other.

Cournot vs. the benchmarks — the numbers

Two firms competing in quantities land *strictly between* monopoly and competition:

$$Q_m = 45 \ < \ Q_{Cou} = 60 \ < \ Q_c = 90, \qquad P_m = 75 \ > \ P_{Cou} = 60 \ > \ P_c = 30.$$

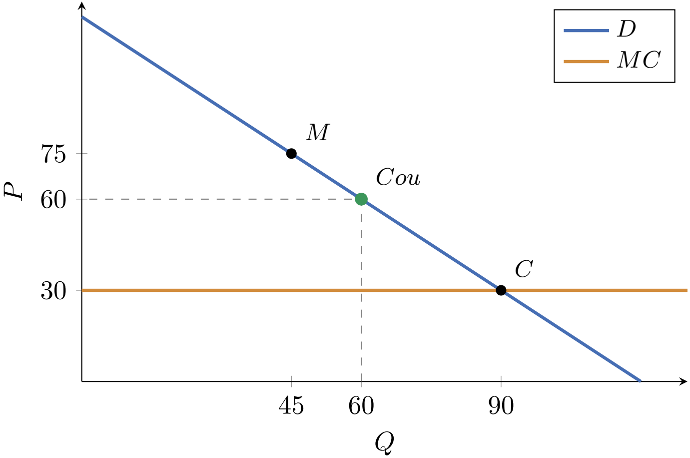
Profit, consumer surplus, welfare

- Each firm: $\pi_i = (60 - 30) \cdot 30 = 900$; industry $\pi = 1800 < 2025$.
- Consumer surplus: $CS = \frac{1}{2} \cdot 60 \cdot 60 = 1800$.
- Total welfare: $W = 1800 + 1800 = 3600$.

Deadweight loss

$DWL_{Cou} = 4050 - 3600 = 450$ — less than a quarter of monopoly's 1012.5. A single quantity-competitor recovers most of the lost surplus, but not all.

Cournot vs. the benchmarks — the picture



Cournot $(60, 60)$ sits between monopoly M and competition C.

Stackelberg — one firm moves first

Same quantity competition, but now the timing changes:

- Firm 1 (the **leader**) chooses q_1 *first*.
- Firm 2 (the **follower**) *observes* q_1, then chooses q_2.

Solve backwards — the follower first

Firm 2 sees q_1 already fixed, so it uses its best response:

$$q_2 = \frac{90 - q_1}{2}.$$

The leader's insight

The leader is not naive: it *knows* this reaction and folds it into its own decision. It does not take q_2 as given — it takes the follower's *rule* as given. This foresight is the whole story.

The leader optimises against the follower's rule

Substitute the follower's response into the leader's demand

$$P = 120 - q_1 - q_2 = 120 - q_1 - \frac{90 - q_1}{2} = 75 - \tfrac{1}{2}q_1.$$

Leader's profit and first-order condition

$$\pi_1 = \left(75 - \tfrac{1}{2}q_1 - 30\right)q_1 = \left(45 - \tfrac{1}{2}q_1\right)q_1, \qquad \frac{d\pi_1}{dq_1} = 45 - q_1 = 0.$$

Stackelberg outcome

$$q_1 = 45, \qquad q_2 = \tfrac{90-45}{2} = 22.5, \qquad Q_{St} = 67.5, \qquad P_{St} = 52.5.$$

Numeric check: the leader produces the monopoly quantity 45; the follower's extra output then pushes price below the monopoly level.

The first-mover advantage

Profits

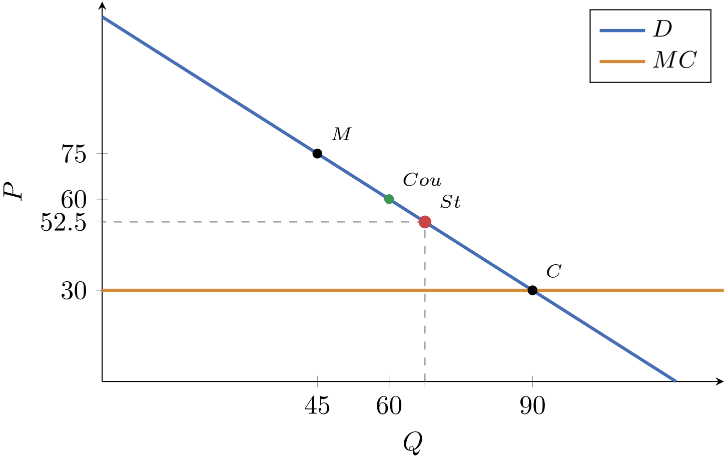
$$\pi_1 = (52.5 - 30) \cdot 45 = 1012.5, \qquad \pi_2 = (52.5 - 30) \cdot 22.5 = 506.25.$$

- The leader earns *more* than a Cournot firm ($1012.5 > 900$).
- The follower earns *less* than a Cournot firm ($506.25 < 900$).
- Total output 67.5 exceeds Cournot's 60, so price is lower and consumers do better still.

Why moving first pays

By committing to a large output *before* the rival chooses, the leader forces the follower to scale back. Commitment, not information, is the source of the advantage.

Stackelberg vs. the benchmarks — the picture



Stackelberg $(67.5, 52.5)$ sits between Cournot and competition — more output, lower price than Cournot.

Four ways to run one market

The whole day on one line

Structure	Q	P	Industry π	CS	Welfare W
Monopoly	45	75	2025	1012.5	3037.5
Cournot (2 firms)	60	60	1800	1800	3600
Stackelberg	67.5	52.5	1518.75	2278.1	3796.9
Bertrand = Perfect comp.	90	30	0	4050	4050

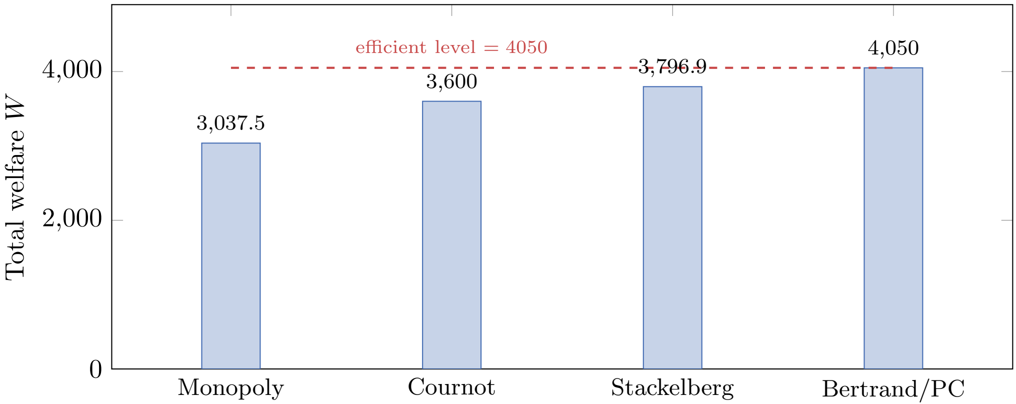
As we move down the table:

- quantity **rises** and price **falls**;
- consumer surplus **rises** throughout, while industry profit **falls**;
- total welfare **rises** — competition is efficient.

Deadweight loss shrinks with competition

$DWL_m = 1012.5 \ > \ DWL_{Cou} = 450 \ > \ DWL_{St} = 253.1 \ > \ DWL_B = 0.$

One picture: welfare across market structures



The gap between each bar and the red line is that structure's deadweight loss. Competition closes the gap entirely.

Day 1 checkpoint

You can now

- derive the **Bertrand** outcome ($P = c$) and explain the two-firm “paradox”;
- find the **Cournot** outcome from best-response functions and their crossing;
- solve **Stackelberg** backwards and explain the first-mover advantage;
- rank all structures by Q, P, profit, consumer surplus and welfare.

The question hiding under today

Every “resting point” was a pair of mutually-best replies — no firm wished to move. **Tomorrow** we name that idea (Nash equilibrium) and build the general toolkit of *game theory* for strategic choice far beyond duopoly.

Time for the in-class exercises.

- **Exercise 1** — a fresh linear market: solve Bertrand, Cournot and Stackelberg, and rank them.
- **Exercise 2** — welfare accounting: consumer surplus, profit and deadweight loss across the structures.

Work in pairs; we reconvene to compare quantities, prices *and* the welfare ranking.